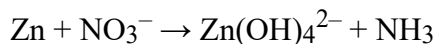


Chapter 20 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. The reaction below occurs in basic solution. In the balanced equation, what is the sum of the coefficients?

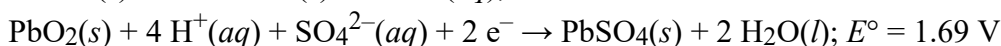
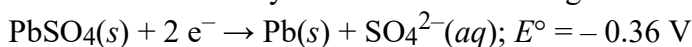


- A) 27 B) 19 C) 15 D) 12 E) 23

2. Which of the following statements is true for a voltaic (galvanic) cell?

- A) The electron flow is from the anode to the cathode.
B) The electron flow is from the positive electrode to the negative electrode.
C) The electron flow is from the negative cathode to the positive anode.
D) The electron flow is through the salt bridge.
E) The electron flow is from the oxidizing agent to the reducing agent through an external circuit.

3. A lead-acid battery involves the following two half-reactions:



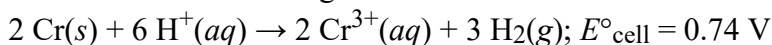
During the discharge reaction of the lead-acid battery at 1.0 M concentrations, the cell potential and the reducing agent are _____ respectively.

- A) 2.05 V and PbO_2
B) 2.05 V and Pb
C) -2.05 V and PbO_2
D) -2.05 V and Pb
E) 1.33 V and Pb

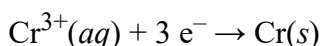
4. What is being reduced at the cathode in the hydrogen fuel cell?

- A) KOH B) Li C) O_2 D) H_2 E) Pt

5. Consider the following cell reaction:



Under standard conditions, what is E° for the following half-reaction?



- A) 0.74 V B) -0.74 V C) 0.37 V D) -0.37 V E) 0.25 V

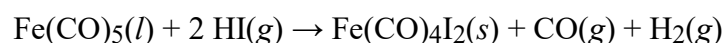
6. What half-reaction occurs at the cathode during the electrolysis of molten potassium bromide?

- A) $\text{K}(s) \rightarrow \text{K}^+(l) + e^-$
B) $\text{Br}_2(l) + 2 e^- \rightarrow 2 \text{Br}^-(l)$
C) $2 \text{Br}^-(l) \rightarrow \text{Br}_2(l) + 2 e^-$
D) $\text{K}^+(l) + e^- \rightarrow \text{K}(s)$
E) $2 \text{H}_2\text{O}(l) + 2 e^- \rightarrow \text{H}_2(g) + 2 \text{OH}^-(l)$

7. The less _____ the value of E°_{red} , the lower the driving force for reduction.

- A) negative B) exothermic C) positive D) endothermic E) extensive

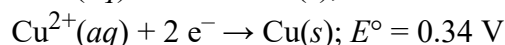
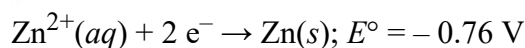
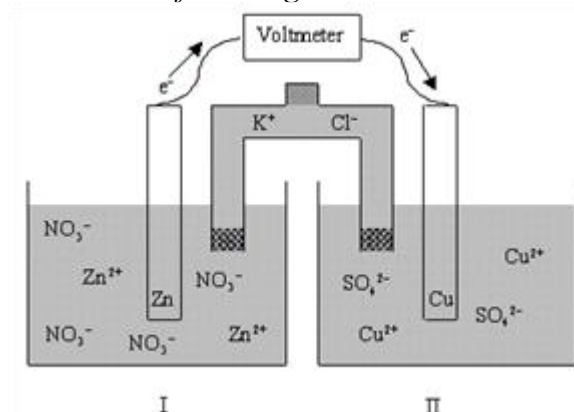
8. Which element is oxidized in the reaction below?



- A) C B) O C) H D) Fe E) I

Chapter 20 Homework

Consider the following voltaic cell to answer the questions 9-11



9. If the cell shown above is initially at standard conditions, which of the following statements is true?

- A) Initially $\Delta G = -212 \text{ kJ}$, and it will not change with time.
- B) Initially $\Delta G = 212 \text{ kJ}$, and it will become more negative with time.
- C) Initially $\Delta G = -212 \text{ kJ}$, and it will become more positive with time.
- D) Initially $\Delta G = 212 \text{ kJ}$, and it will become more positive with time.
- E) Initially $\Delta G = -212 \text{ kJ}$, and it will become more negative with time.

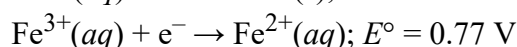
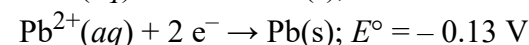
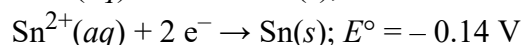
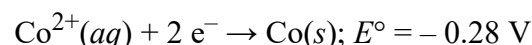
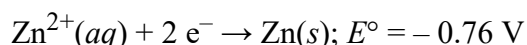
10. If the cell shown above is initially at standard conditions, which of the following statements is true?

- A) Initially $K_c = 2 \times 10^{-37}$, and it decreases with time.
- B) Initially $K_c = 2 \times 10^{37}$, and it does not change with time.
- C) Initially $K_c = 2 \times 10^{37}$, and it decreases with time.
- D) Initially $K_c = 2 \times 10^{37}$, and it increases with time.
- E) Initially $K_c = 2 \times 10^{-37}$, and it increases with time.

11. Which of the following statements is true concerning **half-cell I** as the voltaic cell shown above discharges?

- A) $[\text{Zn}^{2+}]$ increases with time, and $[\text{Cl}^{-}]$ increases with time.
- B) $[\text{Zn}^{2+}]$ decreases with time, and $[\text{Cl}^{-}]$ increases with time.
- C) $[\text{Zn}^{2+}]$ decreases with time, and $[\text{Cl}^{-}]$ decreases with time.
- D) $[\text{Zn}^{2+}]$ decreases with time, and $[\text{NO}_3^{-}]$ increases with time.
- E) $[\text{Zn}^{2+}]$ increases with time, and $[\text{NO}_3^{-}]$ increases with time.

12. Given:

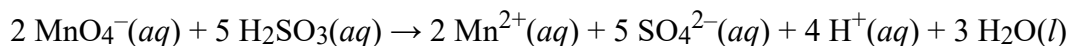


Under standard conditions, which of the following pairs of elements or ions is capable of reducing $\text{Sn}^{2+}(\text{aq})$ to $\text{Sn}(\text{s})$?

- A) $\text{Pb}(\text{s})$ or $\text{Fe}^{2+}(\text{aq})$
- B) $\text{Co}(\text{s})$ or $\text{Pb}(\text{s})$
- C) $\text{Fe}^{2+}(\text{aq})$ or $\text{Zn}(\text{s})$
- D) $\text{Zn}^{2+}(\text{aq})$ or $\text{Co}^{2+}(\text{aq})$
- E) $\text{Zn}(\text{s})$ or $\text{Co}(\text{s})$

Chapter 20 Homework

13. For the cell reaction



the standard cell potential is 1.34 V. Which change(s) will result in an increase in the cell potential?

- I. increasing the permanganate ion concentration
- II. decreasing the pH
- III. increasing the sulfurous acid concentration

A) I only B) II only C) III only D) I and II E) I and III

14. The electrolysis of molten Al_2O_3 for 2.50 hr with an electrical current of 12.0 A produces _____ g of aluminum metal.

A) 90.7 B) 0.373 C) 2.80×10^{-3} D) 10.1 E) 30.2

15. Which of the statements about cathodic protection of a metal pipe against corrosion is correct?

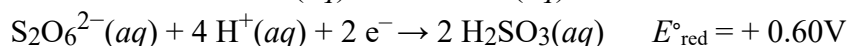
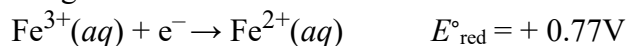
- A) The metal pipe is protected by attaching an active metal to make the pipe the cathode in an electrochemical cell.
- B) The metal pipe is protected by coating the pipe with another metal whose standard reduction potential is less negative than that of the pipe.
- C) The metal pipe is protected by attaching a dry cell to reduce any metal ions which might be formed.
- D) The metal pipe is protected by attaching an active metal to make the pipe the anode in an electrochemical cell.
- E) The metal pipe is protected by coating the pipe with a fluoropolymer to act as a source of fluoride ion (since the latter is so hard to oxidize).

Part B Short questions (15 points)

16. A concentration cell is constructed with two $\text{Zn}(s) - \text{Zn}^{2+}(aq)$ half-cells. In one half-cell $[\text{Zn}^{2+}] = 1.35 \text{ M}$, and in the other $[\text{Zn}^{2+}] = 3.75 \times 10^{-4} \text{ M}$. $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$

- (a) Which half-cell is the anode?
- (b) What is the standard emf of the cell?
- (c) What is the emf of the cell?
- (d) For the anode compartment, predict whether $[\text{Zn}^{2+}]$ will increase, decrease or stay the same as the cell operates.

17. Given the following reduction half-reactions:



- (a) Write balanced chemical equation for the oxidation of $\text{Fe}^{2+}(aq)$ by $\text{S}_2\text{O}_6^{2-}(aq)$.
- (b) Calculate ΔG° for this reaction at 298 K.
- (c) Calculate the equilibrium constant K for this reaction at 298 K.

Chapter 20 Homework Answer Sheet

Name:

Student ID:

Instructor:

Score:

Part A Multiple choice (15 points)

1-5_____

6-10_____

11-15_____

Part B Short questions (15 points)